

Heading group 11

Problem statement / use case definition

Business Problem

Limited insight into copay-card utilization impact on adherence and program cost-efficiency.

Solution

Data set

Concerns:

Synthetic - easier to proceed and process

Not so accurate

Real data - hard to find, need to permission from pharma companies

Plus points : better data , realistic accurate

Data attributes (Copay program transaction data)

Fields/Metrics it contains:

Field	What it means
Transaction ID / Redemption ID	Unique ID per copay use
Card/Coupon ID	Which copay card the patient used
BIN / PCN / Group ID	Routing codes that identify the copay program at the pharmacy system (like an insurance card's routing info)
Transaction date/time	When the card was used
Amount requested vs. amount approved	What the pharmacy asked for vs. what the program paid
Remaining annual/monthly benefit balance	How much of the patient's yearly cap is left

Rejection reason (if declined)	E.g., "benefit exhausted," "not eligible," "duplicate claim"
Pharmacy ID	Which pharmacy processed it
Redemption count	How many times this patient/card has been used (1st fill, 2nd, etc.)

Metrics/KPIs built from this data:

Redemption rate — % of enrolled patients who actually use the card

Average benefit paid per redemption

Cost per redemption — industry benchmarks show roughly \$2.50 per redemption at scale for processing costs [Rx Almanac](#)

Benefit exhaustion rate — % of patients who hit their annual cap

Rejection rate and reasons

Pharmacy-level performance — copay vendors monitor pharmacy behavior over time, creating pharmacy scorecards against critical program metrics to identify outliers and potential misuse [UBC](#)

Accumulator/maximizer exposure — a copay adjustment program ensures only amounts paid by the patient (not the manufacturer coupon) count toward their deductible and out-of-pocket maximum — and coupon use isn't always visible in pharmacy claims systems to anyone but the manufacturer, making this one of the hardest metrics to calculate directly

Data attributes (pharmacy claims)

Core fields/metrics it contains:

Field	What it means
BIN / PCN / Group ID	Routes the claim to the correct processor, plan, and benefit group — incorrect data here causes rejections PillPilot
Cardholder/Patient ID	Insurance member ID
NDC (National Drug Code)	Identifies the exact drug/dosage/manufacturer
Days Supply	How many days the dispensed quantity is meant to cover Ncpdp
Quantity Dispensed	Amount of drug given
Prescriber ID	Which doctor prescribed it
Ingredient Cost / Dispensing Fee	Cost breakdown of the drug itself vs. pharmacy's fee

Date of Service	Fill date
Adjudication response	Paid or Rejected
Reject Code(s)	If rejected, why

Metrics you can calculate from PHARMACY CLAIMS DATASET

- Adherence rate
- Persistency / treatment duration
- Refill frequency
- Days covered
- Treatment gaps
- Discontinuation rate

Metrics you can derive from Copay program transaction dataset

- Copay utilization rate
- Total program spending
- Average assistance per patient
- Average patient out-of-pocket cost
- Program enrollment/participation rate
- Number of successful vs. rejected transactions
- Assistance utilization over time

Fields and metrics you can calculate from PHARMACY CLAIMS DATASET

Core `pharmacy_claims` Table — Fields Needed

Field Name	Type	Description
<code>claim_id</code>	text	Unique ID per claim/transaction
<code>patient_id</code>	text	Tokenized patient identifier

ndc_code	text	National Drug Code (identifies exact drug)
date_of_service	date	The date the drug was actually filled/dispensed
days_supply	integer	How many days this fill is meant to last (e.g., 30, 90)
quantity_dispensed	decimal	Units dispensed (tablets, mL, etc.)
fill_sequence_number	integer	1 = first fill, 2 = first refill, 3 = second refill, etc.
prescriber_id	text	Doctor who prescribed it
pharmacy_id	text	Pharmacy that dispensed it
adjudication_status	text	Paid / Rejected / Reversed
reject_code	text (nullable)	NCPDP reject code if rejected (e.g., 75, 79)
ingredient_cost	decimal	Drug cost
patient_paid_amount	decimal	What patient paid out-of-pocket for this fill
plan_paid_amount	decimal	What insurance paid
expected_next_fill_date	date (derived)	date_of_service + days_supply

Metric-by-Metric: What Fields You Use & How

1. Adherence Rate (usually measured as PDC — Proportion of Days Covered)

Formula: (Total days covered by fills in period) / (Total days in the measurement period)

Fields used: `date_of_service`, `days_supply`, `patient_id`

How: Sum up `days_supply` across all fills in, say, a 365-day window, then divide by 365.

Overlapping days (if patient refills early) are only counted once.

Example: Patient had fills covering 270 days out of 365 → $PDC = 270/365 = 74\%$ adherent.
(Industry standard: $PDC \geq 80\% = \text{"adherent"}$)

2. Persistency / Treatment Duration

Formula: `Last fill date - First fill date` (until a gap signals discontinuation)

Fields used: `date_of_service` (min and max), `patient_id`

How: Find the patient's very first `date_of_service` and their last one (before they stopped refilling). The difference in days = how long they stayed on therapy.

3. Refill Frequency

Formula: `Count of fills / Time period` (e.g., fills per year), or average days between fills

Fields used: `fill_sequence_number`, `date_of_service`, `patient_id`

How: Count total claims per patient in a given period, or calculate the average gap between consecutive `date_of_service` entries for the same patient.

4. Days Covered

Formula: Simply `SUM(days_supply)` for a patient over a period

Fields used: `days_supply`, `date_of_service`, `patient_id`

How: This is the numerator used inside the Adherence Rate/PDC calculation above — but you'd also report it standalone (e.g., "this patient was covered for 245 out of the last 300 days").

5. Treatment Gaps

Formula: `Next fill's date_of_service - (Previous fill's date_of_service + previous days_supply)`

Fields used: `date_of_service`, `days_supply`, `patient_id`, sorted by `fill_sequence_number`

How: For each patient, calculate the `expected_next_fill_date` (previous fill date + days supply). Compare it to the *actual* next `date_of_service`. If the actual fill happened much later than expected → that gap in days is your treatment gap.

Example: Fill 1 on Jan 1, `days_supply` = 30 → expected next fill by Jan 31. Actual next fill was Feb 20 → gap = 20 days.

6. Discontinuation Rate

Formula: % of patients whose gap since last fill exceeds a threshold (commonly 60 or 90 days) with no new fill after

Fields used: `date_of_service` (most recent), `days_supply`, `patient_id`, current/reference date

How: For each patient, check: $(\text{Today's date} - (\text{last fill's date_of_service} + \text{days_supply})) > 60 \text{ days?}$ If yes, and no newer fill exists, flag them as discontinued. Then: **Discontinuation Rate** = $(\# \text{ patients discontinued}) / (\text{Total enrolled patients})$.

Fields & Metrics you can derive from Copay program transaction dataset

Core `copay_transactions` Table — Fields Needed

Field Name	Type	Description
<code>transaction_id</code>	text	Unique ID per copay card usage/redemption
<code>patient_id</code>	text	Tokenized patient identifier
<code>card_id</code>	text	The copay card/coupon ID assigned to the patient
<code>enrollment_date</code>	date	When the patient signed up for the copay program
<code>transaction_date</code>	date	Date the card was used at the pharmacy
<code>pharmacy_id</code>	text	Pharmacy where it was used
<code>full_drug_cost</code>	decimal	Total cost of the drug before any assistance
<code>plan_paid_amount</code>	decimal	What insurance paid

amount_requested	decimal	What the pharmacy asked the copay card to cover
amount_approved	decimal	What the copay card actually paid
patient_paid_amount	decimal	What the patient paid out-of-pocket (after card + insurance)
annual_benefit_cap	decimal	Max the card will pay this patient per year
remaining_balance	decimal	How much of the annual cap is left after this transaction
transaction_status	text	Approved / Rejected / Reversed
rejection_reason	text (nullable)	Why it was rejected, if applicable
total_enrolled_patients	(from patients/enrollment table)	Needed as denominator for participation rate

Metric-by-Metric: What Fields You Use & How

1. Copay Utilization Rate

Formula: (# patients who used the card at least once) / (# patients enrolled in the program)

Fields used: `patient_id` (distinct users in `copay_transactions`), `total_enrolled_patients`

How: Count distinct `patient_ids` with at least one `transaction_status = Approved`, divide by total enrolled patients.

2. Total Program Spending

Formula: `SUM(amount_approved)` across all transactions

Fields used: `amount_approved`

How: Simple sum — this is the total dollar amount the pharma company has paid out through the copay program (e.g., over a month, quarter, or year).

3. Average Assistance Per Patient

Formula: `Total Program Spending / Number of unique patients who received assistance`

Fields used: `amount_approved`, `patient_id`

How: Sum `amount_approved` grouped by `patient_id`, then average across all patients.

4. Average Patient Out-of-Pocket Cost

Formula: `AVG(patient_paid_amount)` across transactions (or per patient, then averaged)

Fields used: `patient_paid_amount`

How: Simple average — tells you how much patients are still paying even *with* the copay card's help (useful for spotting if the assistance is actually enough).

5. Program Enrollment / Participation Rate

Formula: (# patients enrolled in copay program) / (# patients eligible or prescribed the drug)

Fields used: `enrollment_date` (from enrollment table), total prescribed patient count (from your `patients/fills` table)

How: This one technically needs a denominator from *outside* this table — total prescribed patients — so you'd join against your `patients` table to calculate it.

6. Number of Successful vs. Rejected Transactions

Formula: `COUNT(transaction_status = Approved)` vs. `COUNT(transaction_status = Rejected)`

Fields used: `transaction_status`, `rejection_reason`

How: Simple group-by count. Break down rejected ones further by `rejection_reason` (e.g., "benefit exhausted," "not eligible," "duplicate claim") — this tells you *why* transactions are failing, which is very actionable.

7. Assistance Utilization Over Time

Formula: SUM(amount_approved) grouped by month/week

Fields used: transaction_date, amount_approved

How: Group transactions by month (from transaction_date), sum amount_approved per month — gives you a trend line showing whether spending is rising, falling, or seasonal (e.g., spending often spikes in January when deductibles reset).